Greenhouse Effect (Part 1): Heating by Radiation

1. **Calculating the temperature of the Earth (Part II)**

   (a) We found last week in Discussion (and in Section 4.1 of your textbook) that: (i) The Sun emits a total power of about 4×10^{26} W, where a Watt, W, is a $\frac{\text{J}}{\text{s}}$; (ii) The solar flux at the position of the Earth in orbit is about 1400 W/m^2 (this is called the "Solar constant", S); (iii) The average solar flux at the surface of the earth, ignoring albedo, is exactly $F_{\text{Sun at surface (ideal avg)}} = S/4$; (iv) The albedo of the earth is about 0.3. Review each of these four statements with your group and then say a few words about each (what it means, how it was calculated, include a picture if it helps).

   (b) Write down an equation for the average flux of the sun at the surface, $F_{\text{Sun at surface, avg}}$, in terms of S and α (i.e., combining (iii) and (iv) from (a)). Explain why it has this form and calculate a numerical value.

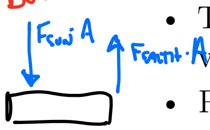
   (c) Repeat the calculation from Q3 of Discussion last week ("Part I") to estimate the Earth's equilibrium temperature. Recall that we did this by assuming that $F_{\text{in}} = F_{\text{out}}$, where the incoming flux is from the sun (just found in (b)) and the outgoing flux is the Earth's blackbody radiation, $F_{\text{Earth surf}}$, which depends on its temperature. Discuss with your group the equilibrium temperature value you find and say a few words about it here.

   (d) Let's construct an "energy flux[1] diagram" associated to the calculation in part (c). Draw a single, wide, rectangular box, representing the surface of the Earth. Draw two flux arrows, one coming into the Earth surface box from above, and one going out of the Earth surface box (also above). Although we had more precise names for these fluxes above, let's now call them F_{sun} and F_{earth} (label your arrows). Since we have assumed a situation of equilibrium, the Earth's total energy is constant, so you can indicate that with a horizontal arrow ($\leftrightarrow$) inside the box. Also write a T_{Earth} inside the box to represent this constant energy value, the equilibrium temperature. We know the value for the incoming arrow; write it next to the arrow. Describe how this picture represents the equation in part (c). What must be the value for the outgoing arrow?

   (e) We will now attempt to correct the value found in (c). Let's start by adding second "object" to our energy flux diagram: the atmosphere.

   - Draw two wide rectangles, one above the other, with some separation. Label the bottom "Earth's Surface" and the top "Atmosphere."
   - The incoming flux arrow to the Earth surface that you drew in (d) remains. Why does it go into the surface box, not[2] the atmosphere box?
   - The outgoing flux arrow from the Earth surface also remains, but it's value might be different in the end. This arrow does not go out into space as before. Why not?
   - The atmosphere should have two flux arrows coming out, each will have the same value, call it F_{atm}. Why does each have the same value?
   - Put "unchanging" arrows ($\leftrightarrow$) inside the boxes. Give a temperature to each box.

CLEAN THIS UP OR JUST USE ENERGY FLOW w/



[1] This is similar to our prior "energy flow diagram" but now the flowing quantity is flux, energy *per area* per time. As you will see in your construction, this will mean that flux is passing from one two-dimensional surface to the next (like the approximately two-dimensional atmosphere above our heads).

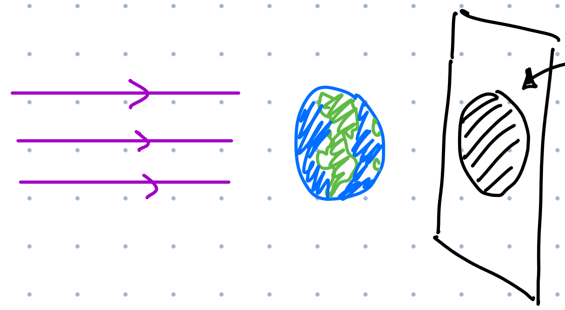
[2] This arrow, from the sun to the Earth could go "behind" the atmosphere box, or around it, but should still come from the top of the page.

1 (a) (i) $\left(T_{SUN} = 5800K \, , \, R_{SUN} = 6.96 \times 10^8 m\right) \xrightarrow{F = \sigma T^4} L_\odot = 4\pi R_\odot^2 F$

$= 3.9 \times 10^{26} W$ (3.8)

(ii) $d_{\odot \to \oplus} = 1.5 \times 10^{11} m \longrightarrow S = F_{SUN @ EARTH} = \dfrac{L_\odot}{4\pi d^2_{\odot \to \oplus}} = 1390 \dfrac{W}{m^2}$ (Book says $1360 W/m^2$)

(iii)



PROJECTION IS $\pi R_\oplus^2$

$\Rightarrow P_{RECEIVED} = S \pi R_\oplus^2$

$F_{SUN, AVG} = \dfrac{P_{RECEIVED}}{4\pi R_\oplus^2}$

$= \dfrac{S \pi R_\oplus^2}{4\pi R_\oplus^2} = \dfrac{S}{4}$

AVG THE POWER OVER THE WHOLE SURFACE

(iv) ALBEDO IS REFLECTIVITY OF EARTH

(BY CLOUDS & ICE & OTHER THINGS).

$$F_{SUN, AVG, ACTUAL} = \frac{S}{4} \cdot (1 - \alpha)$$

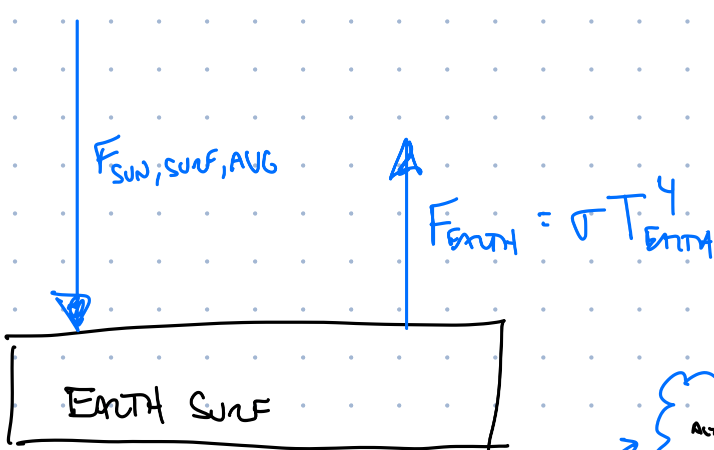
(b)

$$F_{SUN @ SURF. ACT} = \frac{S}{4}\underbrace{(1-\alpha)} = \frac{1390 \, W/m^2}{4}(1 - 0.3)$$

$$= 240 \frac{W}{m^2}$$

WHAT IS NOT REFLECTED IS WHAT ARRIVES!

(c) FOR A PLANET W/ NO ATMOSPHERE



IN EQUILIBRIUM:

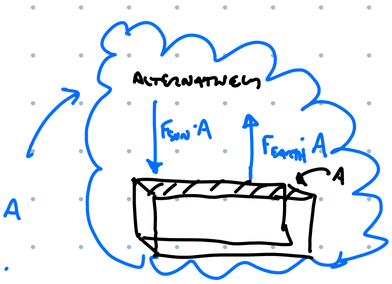
$F_{SUN} = F_{EARTH}$

$240 \frac{W}{m^2} = \sigma T_{EARTH}^4$

$\Rightarrow T_{EARTH} = 255 K$

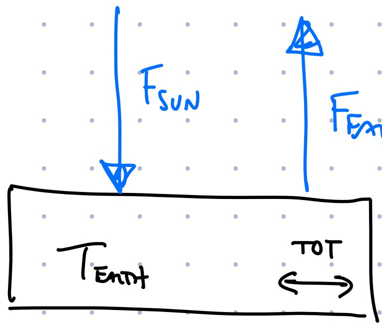
$= -18°C$

THIS "ENERGY FLUX" DIAGRAM IS AN "ENERGY FLOW" DIAGRAM FOR A $1 m^2$ PATCH OF EARTH SURFACE.



TEMPERATURE IS TOO LOW... BUT IS EXACTLY THE VALUE OF TOTAL EARTH.

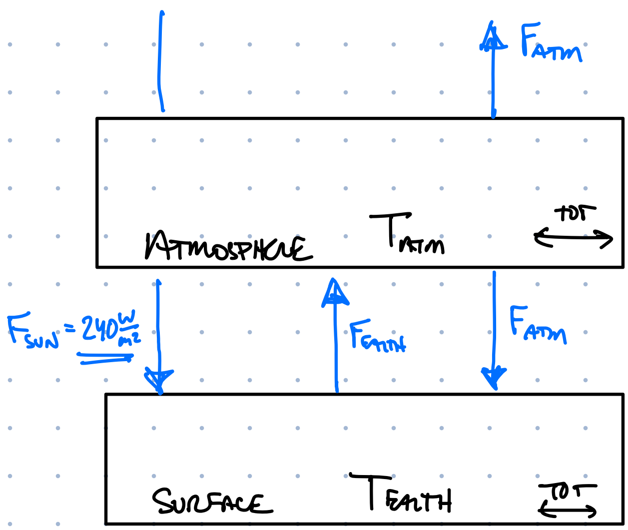
(d)



$F_{EARTH} = F_{SUN} = 240 \frac{W}{m^2}$!

EQUILIBRIUM IMPLIES UNCHANGING ENERGY INTERNAL, UNCHANGING T_{EARTH} AND $F_{EARTH} = F_{SUN}$

(e)



(f) <u>ATMOSPHERE BOX</u>

$$F_{IN} = F_{OUT}$$

$$\boxed{F_{EARTH} = 2\,F_{ATM}}$$

<u>SURFACE BOX</u>

$$\boxed{F_{SUN} + F_{ATM} = F_{EARTH}}$$

$$F_{ATM} = F_{EARTH}/2$$

$$F_{SUN} + \left(F_{EARTH}/2\right) = F_{EARTH}$$

$$\boxed{F_{SUN} = \frac{F_{EARTH}}{2}}$$

(g) NOW WE HAVE

$$F_{SUN} = \frac{F_{EARTH}}{2}$$

$$240 \frac{W}{m^2} = \frac{\sigma T_{EARTH}^4}{2}$$

$$T_{EARTH} = 303\,K = \underline{\underline{30^\circ C}}$$

THIS TEMP. IS TOO HOT ($30^\circ C = 86^\circ F$!). BUT OUR CALC. IS GOING IN THE RIGHT DIRECTION!

- Which flux arrow do we know the number for? Indicate it.

(f) Let's get some equations from your flux diagram. Use the basic principle of $F_{\text{in}} = F_{\text{out}}$ for a box, to get two equations that relate F_{sun}, F_{earth}, and F_{atm} (two boxes means two equations). Get a third equation by combining the two equations to eliminate one variable.

(g) Use your equations, and the Stefan-Boltzmann equation, to get a new temperature estimate for the Earth's surface. Is this one correct?

2. **Measuring solar incident flux:** In Lab this week, we measure first the heat capacity of Aluminum and then use that to measure the radiant flux of a desk lamp (or if we're lucky, the Sun) by tracking the temperature increase of an aluminum block over time.

   (a) We find in lab that the aluminum has a specific heat capacity, c_{Al} about 4.5 times smaller than water (0.90 J/g/K vs. 4.2 J/g/K). Write down the defining equation of the specific heat capacity, and say in your own words what it means. Then describe in plain language what it means that the specific heat capacity of water is 4.5 times higher than aluminum (consider, e.g., 10 g of each substance subjected to the same heating). Can you make sense of the word "capacity" in this context?

   (b) Below is a table of data taken for a class from University of Arizona (where the Sun is a more reliable partner). Let's try to estimate the solar constant[3], S, using the data.

   - Using graph paper, plot this data (Temperature (C) vs. time (s)) being careful to set the range on each axis so that no space is wasted (look at the minimum and maximum values of each quantity).
   - Exam the data and choose a range of times where the temperature seems to be increasing approximately linearly (away from the initial and final "flatness").

BETTER TO SAY TO CHOOSE A REL. SMALL RANGE OF TIMES... AT NO TIME IS IT LINEAR!

Aluminum block dimensions: 7.0 cm x 13.4 cm (93.8 cm²)
Mass of aluminum block: 324 grams
Specific heat of aluminum: 0.215 cal/(gram °C)

Time	Elapsed Time	Temperature
9:09:30	0.0 min	27 C (80.6 F)
9:10:00	0.5	27 (80.6)
9:11:00	1.5	27.5 (81.5)
9:12:00	2.5	28.5 (83.3)
9:13:00	3.5	30.0 (86.0)
9:14:00	4.5	32.0 (89.6)
9:15:00	5.5	33.5 (92.3)
9:17:00	7.5	36.5 (97.7)
9:19:00	9.5	39.5 (103.1)
9:21:30	12.0	42.5 (108.5)
9:25:30	16.0	47.0 (116.6)
9:31:30	22.0	52.0 (125.6)
9:38:00	28.5	56.0 (132.8)

   - What is the average temperature of the aluminum during that time interval?
   - Use a ruler to draw a line of best fit through this portion of the data (you can draw the straight line across your whole graph, but only fit to your chosen data points).
   - Find the slope of that line and its units (always use two widely-separated points *on the line* to find a slope, not data points).
   - The slope you found is $\frac{\Delta T}{\Delta t}$ for the aluminum block. Now use the defining equation for specific heat capacity and the specific heat of aluminum (0.9 J/g/K) to determine the power gain of the aluminum, $\frac{\Delta E}{\Delta t}$.

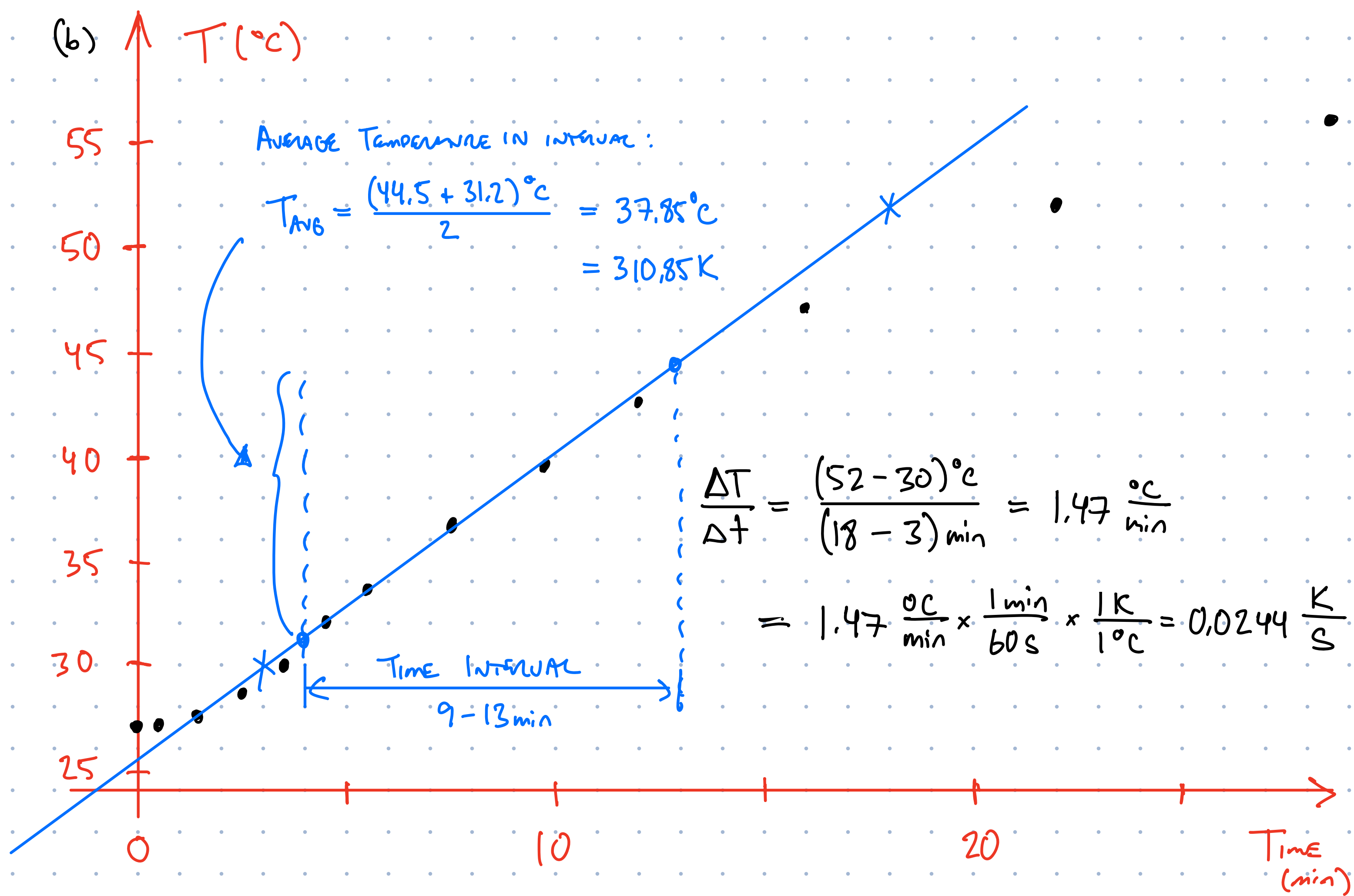
[3] If you point your aluminum block perpendicular to the rays of the sun, and you have a clear sky, then you should be able to estimate S, not the average flux including albedo and the average over the whole Earth's surface!

[2] (a) SPECIFIC HEAT CAPACITY, C, IS THE PROPORTIONALITY CONSTANT BETWEEN THE HEAT TRANSFERRED TO AN OBJECT AND ITS TEMPERATURE INCREASE (PER MASS):

$$Q = c\, m\, \Delta T$$

THE FACT THAT $C_W = 4.5\, C_{Al}$ MEANS THAT FOR THE SAME MASS OF EACH SUBSTANCE, IT REQUIRES 4.5 TIMES AS MUCH ENERGY TO RAISE THE TEMPERATURE OF WATER BY ONE DEGREE THAN IT DOES ALUMINIUM. SIMILARLY, IF YOU HAVE RAISED BOTH SUBSTANCES BY ONE DEGREE THEN WATER MUST HAVE RECEIVED 4.5 TIMES THE ENERGY. THUS WATER'S CAPACITY TO STORE ENERGY (JOULES / °C) IS LARGER.

(b)



$$T_{AVG} = \frac{(44.5 + 31.2)\,°C}{2} = 37.85\,°C$$
$$= 310.85\,K$$

$$\frac{\Delta T}{\Delta t} = \frac{(52 - 30)\,°C}{(18 - 3)\,min} = 1.47\,\frac{°C}{min}$$

$$= 1.47\,\frac{°C}{min} \times \frac{1\,min}{60\,s} \times \frac{1\,K}{1\,°C} = 0.0244\,\frac{K}{s}$$

WE SEE THAT, BETWEEN 9 AND 13 MIN, THE AVERAGE TEMPERATURE AND ITS RATE OF INCREASE ARE

$$T_{AVG} = 37.85\,°C \quad \text{AND} \quad \frac{\Delta T}{\Delta t} = 0.0244\,\frac{K}{s}$$

THE ENERGY INCREASE OF THE ALUMINUM IS FOUND FROM THE FIRST LAW, WHERE NET HEAT INPUT IS ASSOCIATED TO THE TEMPERATURE CHANGE:

$$\Delta E = Q + W = Q + 0 = m c \Delta T$$

$$\Rightarrow \frac{\Delta E}{\Delta t} = C_{Al}\, m_{Al}\, \frac{\Delta T}{\Delta t} = \left(0.9\,\frac{J}{g \cdot K}\right)(324\,g)\left(0.0244\,\frac{K}{s}\right) = 7.16\,\frac{J}{s}$$

- Draw an energy flux diagram for the aluminum block. You should have two[4] flux arrows. In this case, the block is *not* in equilibrium, so you should have an arrow inside the box showing the direction of energy change.
- This diagram implies a non-equilibrium equation of the form:

$$\frac{\Delta E}{\Delta t} = \frac{\Delta E_{\text{in}}}{\Delta t} - \frac{\Delta E_{\text{out}}}{\Delta t}$$
$$\frac{\Delta E}{\Delta t} = F_{\text{block}}\, A_{\text{block}} - F_{\text{sun}}\, A_{\text{block}}$$

  We calculated the left-hand side number from our graph. The block area, A_{block}, can be calculated from the data in the table (convert to meters). We know the value of the flux of the block outward from its average temperature and the Stefan-Boltzmann law. The flux from the sun is our unknown variable; solve for it!

3. **Freezing water in the desert night (BONUS in F2025)**: When it is nighttime, we don't get the Sun's incoming flux. Luckily, all of the objects around us have retained energy throughout the day, and radiate their own infrared light (heat) to keep us relatively warm until sunrise. If, however, you can restrict an object's line of sight to only the sky — and insulate it thermally to minimize conduction from the ground and air — then you will find that it quickly radiates away its heat into the sky!

   (a) You made a number of temperature measurements of the sky using your infrared thermometer. What was the average value you (and your classmates) found? Recall that we have the data results in a google survey from the first weeks of class.

   (b) Using the same equation as the previous question, estimate the rate of energy change, $\frac{\Delta E}{\Delta t}$, of a puddle of water. How long would it take the water to decrease from 20°C to the freezing point?

[4]You could have flux arrows out of the block into the insulating styrofoam, but those will be approximately cancelled by flux arrows into the aluminum from the styrofoam, so we need only include the arrows in/out of the sky-facing surface.

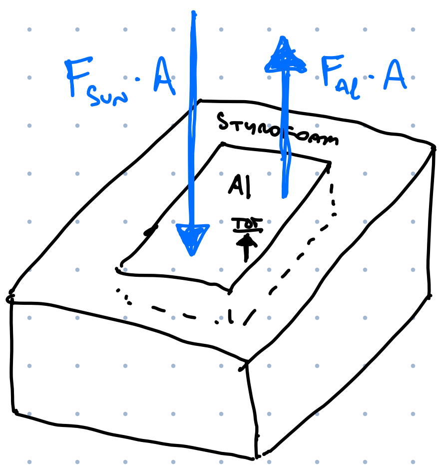
THE ALUMINUM BLOCK IS NOT IN EQUILIBRIUM SO WE HAVE:

$$\Delta E = |E_{IN}| - |E_{OUT}| \qquad \text{NOT} \qquad E_{IN} = E_{OUT}$$

AND FLUX $\left(\frac{\text{ENERGY/TIME}}{\text{AREA}}\right)$ IS RELATED AS:

$$F = \frac{\Delta E}{\Delta t}\Big/ A \quad \Rightarrow \quad \frac{\Delta E}{\Delta t} = F \cdot A$$

SO FOR AN ENERGY FLOW DIAGRAM WE CAN DRAW:



SHOWING THAT THE INCOMING ENERGY FROM THE SUN (OVER SOME TIME INTERVAL Δt) IS LARGER THAN THE BLOCK'S OWN BLACKBODY FLUX INTO THE ENVIRONMENT, SO THE TOTAL ENERGY OF THE BLOCK IS RISING.

THE EQUATION ASSOCIATED TO THIS DIAGRAM IS:

$$\frac{\Delta E}{\Delta t} = F_{SUN} \cdot A - F_{Al} A$$

$$= F_{SUN} A - (\sigma T_{Al}^4) A$$

S-B LAW FOR BLOCK

WHICH WE CAN SOLVE FOR THE SUN'S FLUX:

$$F_{SUN} = \frac{\Delta E}{\Delta t} \cdot \frac{1}{A} + \sigma T_{Al}^4$$

$$= \left(7.16 \ \frac{J}{s}\right) \cdot \frac{1}{9.38 \times 10^{-3} \ m^2} + \left(5.7 \times 10^{-8} \ \frac{J/s}{m^2 K^4}\right) (310.85 \ K)^4$$

$$= 1295.5 \ \frac{J/s}{m^2}$$

$$A_{Block} = 7.0 \ cm \times 13.4 \ cm$$
$$= 93.8 \ cm^2$$
$$= 93.8 \ cm^2 \times \frac{1 m}{100 cm} \times \frac{1 m}{100 cm}$$
$$= 9.38 \times 10^{-3} \ m^2$$

THIS IS REMARKABLY CLOSE TO THE ACTUAL SOLAR CONSTANT!

$$S = 1360 \ \frac{J/s}{m^2}$$

(SEE DESSLER 4.1)

(AND, I THINK, A MORE ACCURATE RESULT THAN THAT OBTAINED IN THE ORIGINAL ANALYSIS AT THE VARIZE WEBSITE, WHERE THE BB RADIATION OF THE Al BLOCK WAS NOT TAKEN INTO ACCOUNT.)